BrandLens

Pre-publication brand governance with rules first, cited AI second, and humans in control.

SUBMISSION READINESS

Feature-complete for the documented Round 2 scope, but not yet judge-ready as a hosted product. A verified live URL and successful production build remain required. No link is represented as live until deployed and smoke-tested.

Project links

	Pending deployment. Earlier Railway URL was not verified and is omitted.
	https://github.com/Utsav-83/BrandLens
	Aurora Pay; one-click creator, manager, reviewer, and admin demo roles when DEMO_MODE=true.

Judging readiness snapshot

Category	Evidence-based status	Readiness
Core workflow (30)	Submission, analysis, decisions, review, approval, audit routes and an E2E smoke script; no live end-to-end run.	Strong code; hosted proof pending
Technical depth (25)	Rules + validated LLM, BM25, Prisma/Postgres, server guards, rate limits, audit chain; build and DB checks blocked locally.	Deploy verification pending
Round 1 progress (20)	Documented pivots from split FastAPI/React, vector retrieval, basic logs, and hard LLM dependency.	Documented; attach original proposal if needed
Round 3 plan (15)	Three substantial improvements with product value and measurable outcomes on page 4.	Ready
Usability (10)	Seeded demo, role logins, sample draft, AI fallback notice, and accessible interaction patterns.	Verify hosted app

Completed features

BrandLens checks drafts against organization-approved sources and routes findings through a human-controlled review workflow.

Ingestion

Text, DOCX, PDF, TXT and Markdown; magic-byte checks and bounded uploads. Digital-text PDFs work; scanned image-only PDFs need OCR.

Rules

Glossary replacements, unsupported absolutes, numeric facts, tone/slang, readability and accessibility-related text checks run without an LLM.

Retrieval and AI

BM25 selects source passages. Provider-neutral LLM calls use environment configuration; citation quotes and proposed numbers are checked against retrieved sources. Failures produce visible rules-only results.

Human workflow

Creators accept, edit or reject findings and submit for review. Managers/reviewers request changes or approve; server-side roles, ownership and separation of duties protect actions.

Audit and operations

PostgreSQL via Prisma 7, DB-backed rate limits, SHA-256 audit chain with Postgres transaction locks, integrity verification, and bounded-batch audit reads.

Judge onboarding

Seeded Aurora Pay sources, four demo roles, sample flawed copy, and explicit loading, empty, error, and AI-unavailable states.

Workspace verification

Vitest: 290 passed, 1 skipped. ESLint: passed. TypeScript no-emit: passed. Production build is unverified: Turbopack worker startup failed in this Windows sandbox; the webpack diagnostic build hit EPERM creating generated .next types. Prisma migration status could not reach the configured database. The PostgreSQL concurrency integration test is skipped without a separate test database.

Technology and implementation

Layer	Implementation
Web	Next.js 16 App Router, React 19, TypeScript, Tailwind CSS 4; Node.js route handlers and server components.
Data	PostgreSQL/Neon target, Prisma 7 custom prisma-client output, @prisma/adapter-pg at runtime; DIRECT_URL for migrations.
Identity	jose HS256 JWT in httpOnly sameSite=lax cookie, bcryptjs, server-side roles and ownership checks.
Review	Deterministic TypeScript rules, BM25 retrieval, fetch-based provider-neutral LLM, 20-second timeout, JSON parsing, source quote validation and rules-only fallback.
Documents	mammoth for DOCX, unpdf for PDF text streams, file signature checks and size limits; image-only OCR is not implemented.
Quality	Vitest unit/security/workflow/retrieval suites, flat-config ESLint, strict TypeScript and 10-step live smoke-test script.

Security controls

Mutating requests validate Origin; API payloads use Zod. Protected routes and server-rendered pages enforce identity and roles; proxy.ts is only a convenience redirect. Draft text and secrets are not intentionally logged. The analysis route has a 60-second max duration. Choose DEMO_MODE deliberately for production and treat demo accounts as public access.

Release blockers

No linked/authenticated Vercel project is available here. The configured database endpoint could not be reached for migration-status verification. Production environment variables, reachable Neon DB, migrations and seeding, successful build, and live smoke test remain release gates.

Changes from the original plan

FastAPI + React to unified Next.js

Reduced cross-origin session and deployment complexity; keeps routes, validation, server rendering and shared types in one TypeScript app.

pgvector to BM25

Prioritized deterministic exact-term retrieval for product names and governance terms, avoiding embedding latency in the core workflow.

Hard LLM dependency to rules-first fallback

Brand checks continue if the provider is absent, fails, or times out; the UI explains that AI review is unavailable.

Basic audit rows to hash chain

Entries link cryptographically and use a PostgreSQL advisory transaction lock for workspace writes.

Visual/OCR review deferred

Round 2 focused on text governance and reliable serverless parsing; scanned/image-only docs are not analyzed.

Three substantial Round 3 improvements

1. Visual creative and accessibility governance

Accept image/design exports, extract text with OCR, inspect logo/color use and contrast, and anchor findings to image regions. Deliver human-reviewed visual checks with labelled evaluation data and measured OCR/contrast accuracy.

2. Governance intelligence across sources

Detect contradictory or superseded facts and guidance, alert on expiring claims, and surface recurring policy errors. Add source lineage and manager resolution actions so teams can repair the knowledge base proactively.

3. Private inference and measurable AI quality

Add local open-model inference (Ollama/vLLM) for sensitive drafts, plus a versioned golden evaluation suite with precision/recall and citation-grounding metrics. Verify private mode makes zero outbound LLM calls.

Before judging

1. Deploy to Vercel with reachable Neon production DB and required variables. 2. Apply migrations and seed demo data. 3. Pass production build and 10-step smoke test against the public URL. 4. Replace pending project link with that verified URL. 5. Confirm the repository link and attach the Round 1 proposal if judges require direct comparison.

Repository: https://github.com/Utsav-83/BrandLens | Hosted project: pending verified deployment